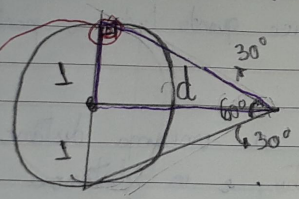
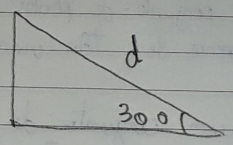
Giovanna Santana Pennisi - CTII 350

Lista de Exercícios - Aula 24

Lista de Exercícios

1- r = 1 / d → distância entre o centro e o vértice





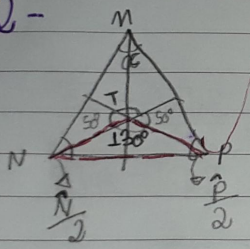
↳ Sen 30° = $\frac{1}{d}$

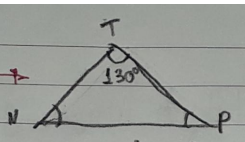
A tangente junto ao raio forma um ângulo reto

$\frac{1}{2} = \frac{1}{d}$ → $d = 2$

Alternativa (d)

2-



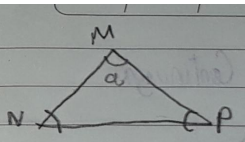


$130° + \frac{\hat{N}}{2} + \frac{\hat{P}}{2} = 180°$

$\left(\frac{\hat{N} + \hat{P}}{2}\right) = 180° - 130°$

$\hat{N} + \hat{P} = 50° \cdot 2$

$\hat{N} + \hat{P} = 100°$



$\hat{N} + \hat{P} + \alpha = 180°$

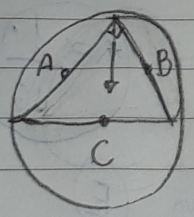
$100° + \alpha = 180°$

$\alpha = 180° - 100°$

$\alpha = 80°$

Alternativa (e)

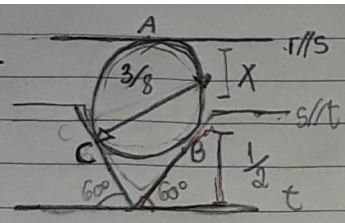
3-



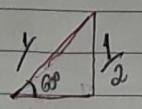
→ Alternativa (B). É retângulo.

↳ O lado do triângulo inscrito é igual ao diâmetro da circunferência. Com isso, o ângulo oposto desse lado é de 90° (um ângulo reto), sendo dessa forma um triângulo retângulo.

4-



→

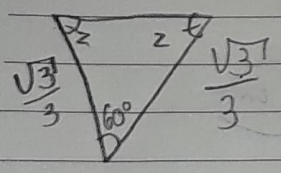


$\text{Sen } 60° = \frac{1}{2} \div y$

→ $\frac{\sqrt{3}}{\cancel{2}} \cdot y = \frac{1}{\cancel{2}}$

$y = \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$

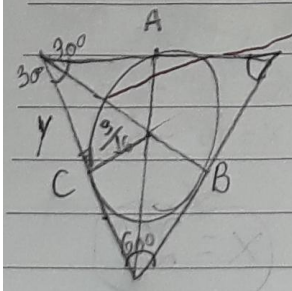
$y = \sqrt{3}/3$



→ $60° + z + z = 180°$

$2z = 180° - 60°$

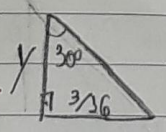
$z = \frac{120°}{3} = z = 60°$ (triângulo equilátero)



$\text{raio} = \frac{3}{8} \div \frac{2}{1}$

$\text{raio} = \frac{3}{8} \cdot \frac{1}{2}$

$\text{raio} = \frac{3}{16}$



$\text{tg } 30° = \frac{3/16}{y}$

$\frac{\sqrt{3}}{3} y = \frac{3}{16}$

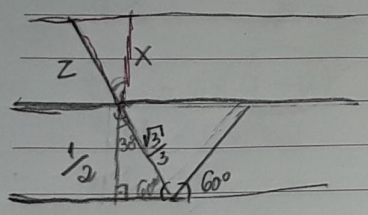
$y = \frac{3}{16} \cdot \frac{3}{\sqrt{3}}$

$y = \frac{9}{16\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$

$y = \frac{9 \cdot \sqrt{3}}{16 \cdot \cancel{3}}$

$y = \frac{3\sqrt{3}}{16}$

4- Continuação



→ Lado do triângulo grande

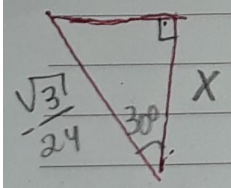
$2 \cdot \frac{3\sqrt{3}}{16 \div 2}$

$\frac{3\sqrt{3}}{8}$

$z = \frac{3\sqrt{3}}{8} - \frac{\sqrt{3}}{3}$

$z = \frac{9\sqrt{3} - 8\sqrt{3}}{24}$

$z = \frac{\sqrt{3}}{24}$



→ $\cos 30° = \frac{x}{\frac{\sqrt{3}}{24}}$

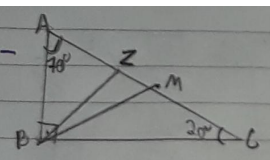
$\frac{\sqrt{3}}{2} = \frac{x}{\frac{\sqrt{3}}{24}}$

$x = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{24}$

$x = \frac{3 \div 3}{48 \div 3}$ → $x = \frac{1}{16}$

Alternativa E

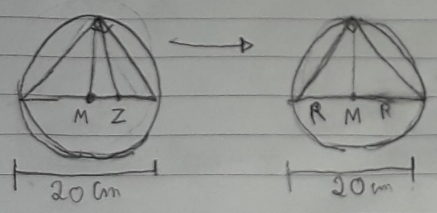
5-



M → mediana

Z → bissetriz.

a)

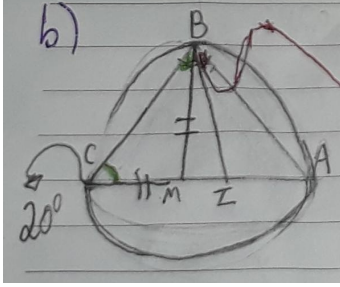


$\rightarrow M = R$

$M = \frac{20}{2}$

$M = 10\ cm$

b)



→ CM = BM = R

→ 45° (BZ é Bissetriz de 90°)

$M\hat{B}Z = X$

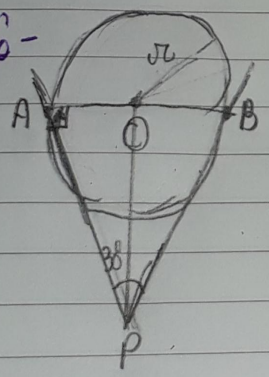
$20° + x + 45° = 90°$

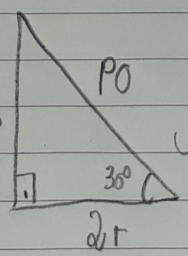
$x = 90° - 20° - 45°$

$x = 90° - 65°$

$x = 25°$

6-





$\text{Sen } 30° = \frac{r}{PO}$

$\frac{1}{2} = \frac{r}{PO}$

Alternativa (C) $PO = 2r$